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SEAT No. :

PD-449

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[6410]-551

T.E. (Mechanical Engineering) (Insem.)

ELECTRIC VEHICLES (Honors)

e-Vehicle System Design

(2019 Pattern) (Semester - II) (302033MJ)

Time : 1Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Two Question from the following (Q1 or Q2, Q3 or Q4.)***
2) ***Draw neat labelled diagram wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Use of non - programmable electronic calculator is permitted.***
5) ***Assume suitable/standard data, if necessary.***

Q1) a) Compare between hydraulic & electric power steering system. **[8]**

b) Explain topology design of bicycle. **[7]**

OR

Q2) a) Explain working principle of Rack & Pinion steering system & Why it is used in Race car? **[8]**

b) Explain two wheeler (2W) configuration of bicycle / dicycle layout. **[7]**

Q3) a) Explain Air Suspension system & its benefits in electric vehicle. **[8]**

b) Explain design of Coil Spring. **[7]**

OR

Q4) a) Discuss the requirements, principles & Components of Suspension System. **[8]**

b) Explain the working of shock absorber with neat sketch. **[7]**

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